

PeerLens v1.0

Complete System Architecture, Algorithmic Engines & End-to-End Operational Guide

Document Version:	1.0 (Production Release)
Target Audience:	Course Instructors, Department Leads, TAs, System Administrators
Core Capabilities:	Intelligent Grouping, 360° Johari Matrix, WebPA Factor Calibration, Realtime Live Projector, Privacy-Safe Portals, Multi-Profile Cloud Synchronization
Supported Platforms:	Desktop Browsers (Chrome, Firefox, Safari, Edge) & Mobile Smartphones
Deployment Model:	Client-Side SPA (Offline-First LocalStorage + Firebase Cloud Sync)

SYSTEM ESSENCE & PHILOSOPHY

PeerLens is built to eliminate the logistical nightmare and mathematical unfairness of group project grading. By fusing advanced Simulated Annealing team formation algorithms, psychometric Johari Window self/peer discrepancy analysis, and WebPA individual score calibration, PeerLens guarantees both diverse collaborative teams and rigorously fair academic evaluations.

Table of Contents

Chapter 1: System Architecture & Core Philosophy	Page 2
Chapter 2: Workspace Profiles & Cloud Authentication	Page 3
Chapter 3: Student Roster Onboarding & Schema Mapping	Page 4
Chapter 4: AutoGroup Studio & Multi-Criteria Optimization Engine	Page 5
Chapter 5: Custom Grading Scales & Rubric Calibration	Page 6
Chapter 6: Link Dispatcher & Automated Email Notification Suite	Page 7
Chapter 7: Live Classroom Presentation Mode (Privacy-Safe Projector)	Page 8
Chapter 8: Student Evaluation Portal Experience	Page 9
Chapter 9: Advanced Analytics: Radar, Johari Window & NLP Themes	Page 10
Chapter 10: The WebPA Mathematical Algorithm & Fairness Calibration	Page 11
Chapter 11: Collusion Audit, Sprints & Student Dossier Exports	Page 12
Chapter 12: Cloud Deployment, Zero-Config Hosting & FAQs	Page 13

1. System Architecture & Core Philosophy

PeerLens is an enterprise-grade academic peer evaluation and algorithmic team-building platform designed for higher education and intensive corporate bootcamps. It addresses the traditional vulnerabilities of collaborative assessments: free-rider inflation, personality bias, uncalibrated raw scoring, and tedious administrative overhead.

Core Technical Pillars:

- **Client-Side High Performance:** Built on React 18, TypeScript, and Vite. All algorithms—including Simulated Annealing and WebPA matrix transformations—execute in real time client-side with zero latency.
- **Dual-Mode Persistence (Offline-First + Cloud Sync):** Seamlessly runs offline using structured localStorage cache. When connected to Firebase Firestore, it provides instantaneous multi-device synchronization and team roster persistence.
- **Single-Blind Privacy-First Evaluation:** Students complete peer ratings and constructive feedback in private verified sessions. PeerLens automatically sanitizes reviewer identities before generating student dossiers.
- **Mobile-First Adaptive Design:** Dedicated responsive layouts ensure seamless usability across instructor smartphones, projection screens, and student mobile devices.

2. Workspace Profiles & Cloud Authentication

PeerLens provides multi-tenant isolation at the browser and cloud level through **Admin Workspace Profiles**. This allows instructors to manage multiple distinct courses, departments, or academic terms independently without data pollution.

Workspace Profile Architecture:

- **Default Profile:** The primary local workspace created automatically upon initial launch.
- **Custom Profiles:** Instructors can click the profile switcher to create isolated profiles (e.g., 'CS101-Fall2026', 'MBA-CohortB'). Each profile maintains its own classes, rosters, rubrics, and historical snapshots.
- **Profile Deletion Safeguards:** Built-in deletion confirmation modals prevent accidental removal of active academic workspaces.

Firebase Cloud Synchronization & Auth:

The Cloud Sync tab enables persistent multi-device collaboration. Connecting Firebase provides secure email/password authentication, real-time Firestore listeners, and multi-instructor access. Even if browser cache is cleared, all rosters, ratings, and milestones remain safely backed up in the cloud.

PRO TIP: MULTI-INSTRUCTOR COLLABORATION

Multiple TAs and professors can share the same Firebase configuration. Any student evaluation submitted from the mobile portal updates the Firestore database and reflects on the instructor's live dashboard instantly.

3. Student Roster Onboarding & Schema Mapping

PeerLens accommodates any existing university roster format through four flexible onboarding channels:

Channel	Description	Ideal Use Case
3-Step CSV Wizard	Intelligent CSV parser that auto-detects column headers (Name, Email, Student ID, Major, Gender, Skills) with visual preview.	Exported rosters from Canvas, Blackboard, Moodle, or Excel.
Self-Enrollment Portal	Generates a QR code and shareable URL where students self-register their details, degree, and skills.	First day of class / in-person auditorium onboarding.
Manual Entry Modal	Direct instructor form for adding individual students or making roster adjustments on the fly.	Late additions, drop-ins, and student profile edits.
Sample Cohort	1-Click generator that injects 24 pre-calibrated engineering students across diverse teams.	Feature testing, live demos, and rubric validation.

4. AutoGroup Studio & Multi-Criteria Optimization Engine

Creating balanced student teams manually is notoriously prone to social clustering and skill polarization. The **AutoGroup Studio** employs a multi-objective **Simulated Annealing Optimization Algorithm** to synthesize maximally effective teams.

Algorithmic Diversity Dimensions:

1. **Skill Stratification:** Distributes high, medium, and developing technical skill levels evenly across all squads.
2. **Academic / Major Heterogeneity:** Mixes complementary degrees (e.g., Computer Science, UI/UX Design, Data Science, Business).
3. **Gender & Demographic Balance:** Ensures equitable representation and prevents isolating minority students.
4. **Global / Nationality Mixing:** Encourages cross-cultural collaboration.

Studio Controls & Features:

- **Target Team Size Input:** Set exact team capacity (e.g., 4 or 5 students per team) with instant automatic squad count calculations.
- **Diversity Strategy Presets:** Choose between '*Maximum Diversity*', '*Skill Balanced*', '*Gender Balanced*', '*Academic Blend*', or '*Pure Random*'.
- **Dynamic Team Naming Styles:** Automatically styles team designations using Mythological Gods, Tech Giants, Constellations, Elements, or Alpha-Numeric codes.
- **Real-Time Diversity Score Ribbon:** Live metric indicators compute real-time scores (0-100%) for Skill Variance, Gender Parity, and Major Distribution.

5. Custom Grading Scales & Rubric Calibration

PeerLens eliminates rigid grading scales by allowing instructors to define completely custom multi-criteria rubrics.

Rubric Customization Features:

- **Dynamic Criteria Creation:** Add unlimited evaluation dimensions (e.g., '*Technical Contribution*', '*Communication & Dependability*', '*Problem Solving*', '*Teamwork Ethic*').
- **Custom Maximum Scale:** Configure each criterion's raw scale (e.g., 1-5 stars, 1-10 points, or 1-100 percentage points).
- **Target Scaled Grade Mapping:** Automatically normalize raw student scores into official course gradebook weights (e.g., scale a 40-point rubric into 100% or a 20-point assignment).
- **Strict Deadline Enforcement:** Set exact dates and times for evaluation cutoff. Expired evaluations lock automatically on the student portal.

6. Link Dispatcher & Automated Email Notification Suite

Distributing unique evaluation links to dozens or hundreds of students is streamlined via the **Link Dispatcher Hub**.

Dispatch Modes & Adapters:

1. **1-Click Bulk Copy Table:** Instantly copy all student names, emails, and direct portal links formatted for spreadsheet pasting.
2. **Demo Mode Simulator:** Test link delivery and verify templates without transmitting actual emails.
3. **EmailJS Client Adapter:** Connect your custom EmailJS service ID and template ID for direct browser-initiated dispatches.
4. **Brevo (Sendinblue) API Integration:** Built-in support for Brevo's high-deliverability transactional email API with 300 free emails/day.

Dynamic Template Tags:

Instructors can customize the notification body with smart merge tags: `{{student_name}}`, `{{class_name}}`, `{{portal_link}}`, and `{{deadline}}`.

7. Live Classroom Presentation Mode (Privacy-Safe Projector)

The **Live Classroom Projector** is an interactive, light-themed presentation tool designed to be cast onto auditorium screens during in-class peer review workshops.

Privacy-Safe Architectural Design:

- **Zero Rating Exposure:** Absolutely no individual grades, scores, or review comments are ever displayed.
- **Team Roster Privacy:** Shows team progress rings and submission completion percentages, incentivizing peer accountability without public shaming.
- **Dynamic 2-Tier Layout:** Seamlessly expands as new teams are added, utilizing dynamic responsive grids.
- **Auto-Refresh Pulse:** Automatically polls the latest cloud submissions and updates progress rings live in real time.

8. Student Evaluation Portal Experience

Students access their private portal using a unique verified token link (e.g., `/?classId=...&studentId=...`). The portal is designed for high usability on smartphones and laptops.

Step-by-Step Student Workflow:

- 1. Identity Confirmation:** The student is greeted with their verified identity, squad designation, and deadline banner.
- 2. Self-Assessment:** The student first scores their own performance across all rubric criteria and provides self-reflection.
- 3. Peer Rating Tabs:** Dedicated interactive rating tabs for each teammate. Sliders provide fluid score selection with live score badges.
- 4. Constructive Feedback Validation:** Students provide written constructive comments for each teammate. The system validates meaningful input length to prevent low-effort spam.
- 5. Cryptographic Submission Lock:** Once submitted, the portal transitions to a verified locked state, preventing post-deadline tampering.

9. Advanced Analytics: Radar, Johari Window & NLP Themes

PeerLens translates raw evaluations into actionable pedagogical insights across multiple psychometric frameworks:

1. 360° Competency Radar Visualizer:

A multi-axis spider radar chart that overlays a student's **Self-Rating** directly against the **Peer-Average Rating** across every rubric criterion.

2. Johari Window Psychometric Matrix:

Quadrant	Self / Peer Profile	Pedagogical Interpretation
Arena (Open)	High Self + High Peer	High-performing collaborative leader with accurate self-awareness.
Blind Spot	Low Self + High Peer	High achiever with imposter syndrome; peers value them higher than they value themselves.
Façade (Hidden)	High Self + Low Peer	Overestimating contribution; potential free-rider or communication friction.
Unknown Arena	Low Self + Low Peer	Disengaged student requiring instructor mentorship and skill development.

3. AI Feedback NLP Sentiment & Thematic Engine:

Categorizes qualitative peer comments into Sentiment Breakdown (Positive, Neutral, Constructive) and clusters them into 4 core thematic pillars: *Technical Quality*, *Communication*, *Collaboration*, and *Leadership*.

10. The WebPA Mathematical Algorithm & Fairness Calibration

The WebPA (Web Peer Assessment) algorithm is the internationally recognized gold standard for calculating individual contributions in group coursework.

Mathematical Formulation & Step-by-Step Derivation:

WEBPA EQUATION BREAKDOWN

Step 1: Normalization of Peer Scores Given by Reviewer j:

Let S_{ij} be the total score given by student j to student i across all criteria.

Normalized Score: $N_{ij} = S_{ij} / (\text{Sum of all } S_{kj} \text{ across all team members } k)$

Step 2: WebPA Fractional Factor (F_i):

Sum all normalized scores received by student i from peers and self, divided by total reviewers R_i :

$F_i = (\text{Sum of } N_{ij}) / R_i$

Step 3: WebPA Individual Multiplier (W_i):

Let T be total team size. The calibrated individual weighting factor is:

$W_i = F_i * T$

Step 4: Final Adjusted Grade:

$\text{Final_Grade}_i = \text{Base_Group_Grade} * W_i$ (Subject to Floor and Ceiling Safety Caps)

Safety Caps & Anti-Penalization Boundaries:

- **Baseline Multiplier:** An average contributor receives a WebPA multiplier of exactly **1.00**.
- **High Performer Ceiling:** Exceptional contributors receive multipliers up to **1.25x** (25% boost).
- **Low Contribution Floor:** Disengaged students are capped at safe lower thresholds to prevent mathematical grade inversion.

11. Collusion Audit, Sprints & Student Dossier Exports

1. Collusion & Outlier Anomaly Audit:

- **Reciprocal Collusion Detection:** Flags student pairs who trade maximum scores while giving lower marks to others.
- **Harsh / Generous Grader Auditing:** Flags graders whose variance deviates significantly from the class distribution.

2. Milestone Sprints & Multi-Period Archival:

Archive evaluations across Project Sprints (e.g., Sprint 1, Midterm, Final Demo). Allows instructors to track longitudinal growth and group dynamic evolution.

3. Exporting & Comprehensive Student Dossiers:

- **1-Click PDF Dossiers:** Generates individualized student PDF reports containing their 360° radar chart, Johari placement, WebPA score, and sanitized constructive feedback.
- **Full CSV Gradebook Export:** Exports student IDs, group names, raw peer scores, WebPA factors, and final scaled grades ready for Canvas, Blackboard, or Excel.

12. Cloud Deployment, Zero-Config Hosting & FAQs

Netlify & Production Hosting:

- **Zero-Config SPA Build:** Pre-configured with netlify.toml and public/_redirects for instant Git-connected continuous deployment.
- **Zero Environment Variable Requirements:** Completely standalone client-side architecture; no server environment keys needed.

Frequently Asked Questions (FAQ):

Q: What happens if a student forgets to submit their peer reviews?

A: The WebPA engine dynamically normalizes the scores from participating reviewers so the remaining group members are not penalized. The non-submitting student is clearly flagged in the instructor's audit deck.

Q: Can students see who wrote which comment?

A: No. PeerLens enforces single-blind anonymity. Individual ratings and comments are aggregated and sanitized in the student dossier.

Q: Does PeerLens work without an internet connection?

A: Yes! PeerLens is offline-first. All data persists in local browser storage, and syncs automatically when cloud connectivity is restored.